

Cloud Computing Unit 2

Virtualization allows multiple operating systems to share a single physical server. A hypervisor is the software layer that creates and manages virtual machines on the host hardware.

Type 1 hypervisors run directly on the physical hardware, which gives better performance than hosted designs. Containers share the host kernel and start much faster than full virtual machines.

Infrastructure as a Service provides virtual machines, storage and networking on demand. Platform as a Service gives developers a managed runtime for deploying applications without handling servers.

Elasticity is the ability of a cloud platform to automatically add or remove resources as the workload changes. Load balancers distribute incoming requests across several servers to improve availability.

Object storage keeps data as objects with metadata and a unique identifier, which makes it ideal for backups and media files. Block storage presents raw volumes to virtual machines for databases.

Service level agreements define the uptime and support guarantees that a cloud provider promises to its customers. Multi-tenancy means several customers share the same physical infrastructure while their data stays isolated.